

Deep Learning for Structured Data – Exercise Sheet 3

Submit the solution in a single PDF in the Moodle until 05.06.2026, 12:00. The corresponding exercise session will take place on 12.06.2026 at 08:30.

Problem 1: (2 Pt.)

Implement the GRU layer from the lecture and benchmark it against a sequence prediction task of your choice regarding

- prediction quality, and
- prediction time.

Can you implement the GRU architecture so that it is much faster than the LSTM?

Problem 2: (4 Pt.)

Design a new attention mechanism that does not rely on dot-product selection attention. Your attention mechanism should still output a row-wise stochastic score matrix. Integrate your newly designed attention mechanism into the transformer layer and benchmark it on a sequence task of your choice against a standard transformer layer in terms of prediction quality.

Problem 3: (3 Pt.)

Design a sequence learning task where an LSTM-based architecture performs better than an RNN. Validate your task empirically, showing that the LSTM-based architecture performs significantly better than the RNN.

Problem 4: (3 Pt.)

Design a sequence learning task where a transformer architecture performs better than an LSTM-based architecture. Validate your task empirically, showing that the transformer architecture performs significantly better than the LSTM-based architecture.

Problem 5: (2 Pt.)

The lecture defines the sequence space as

$$\mathcal{X}_{\text{seq}}^{(d)} := \bigcup_{T=1}^{\infty} (\mathbb{R}^d)^T.$$

A set-based model is permutation-invariant if

$$g(x_1, \dots, x_T) = g(x_{\pi(1)}, \dots, x_{\pi(T)})$$

for every permutation π of $\{1, \dots, T\}$.

Construct a supervised sequence-classification problem

$$f: \mathcal{X}_{\text{seq}}^{(1)} \rightarrow \{0, 1\}$$

that cannot be meaningfully represented by any permutation-invariant model.

Problem 6: (2 Pt.)

In dot-product attention, the attention scores are computed as

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}.$$

Let

$$\mathbf{q}, \mathbf{k} \in \mathbb{R}^{d_k}$$

be two random vectors. Assume that all entries of $\mathbf{q}$ and $\mathbf{k}$ are independent random variables with

$$\mathbb{E}[q_j] = 0, \quad \mathbb{E}[k_j] = 0, \quad \text{Var}(q_j) = 1, \quad \text{Var}(k_j) = 1.$$

Show that

$$\text{Var}(\mathbf{q}^\top \mathbf{k}) = d_k.$$

Then explain why the scaled score

$$\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$$

has variance approximately independent of d_k .

Problem 7: (2 Pt.)

Consider a sequence of length T , model dimension d , and L layers. Assume that all hidden dimensions are of order d . Compare the asymptotic computational cost and the maximum information path length of

1. an RNN,
2. a Transformer encoder with full self-attention.

In particular, discuss the following points:

1. the cost per layer,
2. whether the model can process positions in parallel,
3. the maximum path length between two distant sequence positions.